



# **LAS 6292: Data Collection and Management**

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# Introduction

This book is a manual for students in LAS 6292: Data Collection & Management. The course is designed for graduate students from any discipline – social sciences, humanities, biophysical sciences – and at all stages of their graduate program. It is an introduction to methods for collecting, organizing, managing, and visualizing both qualitative and quantitative data. Students will gain hands-on experience with best practices and tools.

- 🗨 Describe the different types of research data & the research data life-cycle
- ❓ Explain the need for and benefits of data management and sharing
- 📁 Describe and implement best practices for data collection, storage, management, and sharing
- 🔍 Find, download, and analyze publicly available data from repositories
- ✔ Carry out simple and reproducible data corrections and dataset organization
- 🏛 Describe public policies and agency requirements for data management and sharing
- ⚖ Articulate the major legal & ethical issues regarding the data collection, use, and storage
- 📁 Create and Implement Data Management Plans in funder-specific formats
- ✂ Identify and properly use tools for more efficient and secure data collection in the field

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## 💡 Citation

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This guide is a [Quarto Book](#) hosted on the [Bruna Lab's Github site](#). Please suggest edits to the text, missing resources, or make suggestions for improvement either by [pull request](#) or by [posting an issue](#) on the course book's repository. For more information and tutorials for doing so, see [?@sec-manual](#).

**Part I.**

## **Course Topics**

# 1. Naming Conventions

## 1.1. Introduction

The first (and easiest) way to get started organizing your data is to use a simple, clear, and consistent system for naming your files (i.e., a “naming convention”). Using a well-thought out naming convention has a number of important benefits:

- Files will be easier to find
- You won't have to open files to see what they are
- Files are easier to sort
- Files are easier to share with collaborators (and for collaborators to use)
- It helps prevent accidentally overwriting or deleting files

## 1.2. How to name your files

Your file names should:

1. **Tell you about the contents of the file and allow you to uniquely identify it.** For instance (compare 'data 1998' vs 'survey data' vs 'survey data 1998' vs 'small mammal survey data 1998'). You can use things like the an acronym for the project, abbreviation for the study, location where collected, the name of the investigator, the year or month when collected, the type of data type, etc.
2. **Be as short as possible.** Many database systems (like MS One Drive) have limits on file name length. Use a max of 25 characters, and aim for less than that.
3. **Avoid using special characters and spaces** Don't use characters like \$ % ^ & # | :. This is really important because software, automated data processing tools, web browsers, and computer operating systems use spaces and special characters for dividing up (parsing) strings of text, as well as other processes.
4. **Start with letters, never numbers.** Using numbers at the start of file names can lead to problems when they are read into stats software and other programs. Instead of '2020\_survey\_responses' use 'survey\_responses\_2020'
5. **Help you quickly sort files chronologically or numerically.** Dates are very useful in file names because they can help identify the most recent files easily. You can *not* rely on the file date associated with the file on the computer! However, it is really important to use a clear format for dates, especially if collaborating internationally. I learned the hard way that not everyone reads 'census\_data\_3-4-2018' the same way I do, so I started using 'census\_data\_4march2018'. However, even this isn't ideal because of

## 1. Naming Conventions

how it sorts things in a folder. Now I now use this format: 'file\_name\_YYYYMMDD' ('census\_data\_20180403'). You can also use file\_name\_YYYY-MM-DD. This is based on the international format for dates and times ([ISO 8601](#)). In addition to the benefit of sorting more easily, you'll never have to use the the dreaded '\_\_\_\_\_.final.docx' and '\_\_\_\_\_.actual\_final.docx' in your file names ever again...you'll always know which is the last one. If the files don't require a date, but you want to maintain a sequence, use leading zeros to maintain sort order. 7 and 701 don't sort the way you expect them to, so use 007 and 701.

6. **Use a consistent method of dealing with spaces and letter case.** . For a the spreadsheet with responses to a survey by 5 different people, don't use 'survey responses Florida'. Instead use (i) 'survey\_responses\_florida' or (ii) 'SurveyResponsesFlorida'. Try to avoid mixing the two (e.g., 'survey\_responses\_Florida'). By the way, (i) is known as *pothole* and (ii) is known as *camel*. Personally, I prefer *pothole* name is better because of it is easier to read and you never to remember what words get capitalized (because none of the do). Use underscore ( \_ ) or dashes ( - ) to separate meaningful parts of file names.

**Once you have a system, write it down, print it out, and put it up somewhere you can easily refer to it.** Remember -whenever possible make the names simple, informative, and unique. If you collected survey data and behavioral data in 2017, don't call them both `data_2017` even if they are in different folders. Call one '`behavior_obs_2017`' and the other '`survey_responses_2017`'.

Simple names also help you avoid problems with “system and software portability” - you might be working on a Mac but your collaborators are working on Linux or Windows, and the software made for each can read file names in different ways. File names that are simple, lower-case, and have no special characters are less software- and platform-dependent.

## 1.3. Tools & Resources

### 1.3.1. Best Practices & Conventions for Naming Files

- [Smithsonian Library](#)
- [UF Library](#) (*bottom of the page*)
- [Stanford Library](#)

### 1.3.2. Tutorials and Tools for Bulk Renaming of Files

- Batch Renaming of Files in Windows: [Post](#)
- Batch Renaming of Files in MacOS: [Post 1](#), [Post 2](#)
- [Adobe Bridge](#) (*a powerful “asset manager” used to organize collections and edit metadata*)
- [Renamer \(macOS\)](#)
- [Renamer Lite](#)
- [PSRenamer](#)
- [Bulk Rename \(Windows\)](#)

## *1. Naming Conventions*

- [Rename Master \(Windows\)](#)
- [Ant Renamer](#)



## 2. File Organization

### 2.1. File Organization - Electronic

There aren't really "rules" about how to organize your files, but I think the best way to organize is to think in terms of "projects". Of course, how you define a "project" is sometimes tough...is the PhD the 'Project' or is each chapter of the dissertation a 'Project'? You'll have to give this some thought and think of a system that works for you. That said...

**I like 'Project' based organization for two main reasons:**

1. **First**, it pushes you to think in terms of outcomes: 'projects' lead to 'products' (e.g., paper, report, grant proposal). This helps you think more clearly about what data need to be collected and the analyses that need to be done. If the data aren't for a new project, and it's not clear to you if or how the data fit into a pre-existing project, then are you sure you need to spend time, effort, and money collecting them?
2. **Second**, project-based organization focuses on the individual steps and components that take a project from start-to-finish: planning your data collection, data gathering and clean-up, analysis, presentation, and data archiving.

### 2.2. Example: Project Based Organization

**A project-based organization scheme for a master's thesis might look something like this:**

1. Top Folder: `emb_masters_thesis`
2. Sub-folder 1: `Chapter1_lit_review`
3. Nested Folders and Subfolders for:
  - Plans for Data Collection & Management
    - DMP
    - plans for storage and backup
  - data
    - raw data
    - clean data
    - metadata
  - analyses
    - intermediate outputs from analyses
    - final outputs from analyses
  - presentation (e.g., `masters_ch1_text` or `ch1_slides`)

## 2. File Organization

- data archiving

You can have one of these for each thesis chapter. Note that some (DMP, plans for storage and backup) might be better placed one folder up in the `emb_master_thesis` folder

*But what if I will be using the same data for multiple chapters or papers?* This is a great questions, and there are several options. One would be to include both chapters in the same folder, each with its own subfolder (manuscript 1, manuscript 2). The other would be to have each as a separate project, and pull the data in from the online archive where you deposited the data (or folder where the data are stored). As you can probably guess, I think #2 is better because I think of each paper as a different project.

### 2.3. File Organization - Physical

**Whenever possible I try to organize physical records and archives in the same way as digital ones:** in a single box, file folder, cabinet, etc.\*\* try to set it up so that you go to one place for everything related to a project. Of course, you can't always do that because specimens go to museums or stay in an archive...but you can keep the information on how to *find* the originals there. Don't forget to include copies of the survey, data sheets, etc., as well as maps or other such resources.

**Correspondence is a bit more complicated.** You can keep notes - electronic or paper - in the relevant folder. But one thing that has gotten complicated is email correspondence regarding projects, manuscripts, etc. Printing seems like a huge waste of paper, and yet if critical it might be worth doing. Other wise keep an email folder for each project as well - with the same name as the electronic version so you can easily find email correspondence related to each project.

### 2.4. Documentation

**Write it down.** Prepare a file documenting your data organization structure so you can share it with team members (or look at it years later when looking for something). These can be '`README.txt`' files as needed for each folder. The '`README.txt`' file might include such things as explanations of naming conventions and how the structure of the directory relates to the structure of the project.

**There are some excellent tools for organizing projects.** Use appropriate tools, such as version control tools or electronic lab notebooks, to keep track of the history of the data files (more on this later in the semester).

## 3. Data Storage & Backup

### 3.1. What is backup?

#### 1. What is the difference between a backup and an archive?

- Original Data
- Operation Data
- Backup
- Archive

#### 2. Why backup?

- Hardware failure
- Software or media obsolescence
- Virus infection, malware, or malicious hacking
- Power failure
- Human error
- Future validation of results
- Data requests
- Legal Obligations: Lawsuits, FOIA reequests, agency mandates
- Required by funding agency or foundation
- Server, hardware, or software failure
- Financial instability of the projecct or organization

### The Back-up Rule: 3-2-1

**3-2-1:** A secure backup requires a *minimum* of **3 copies** of your data on **2 types of storage media** with **one copy off-site**. Having 1 copy off-site protects your data from local risks like theft, lab fires, flooding, or natural disasters. Using 2 storage media improves the likelihood that at least one version will be readable in the future should one media type become obsolete or degrade unexpectedly. Having 3 copies helps ensure that your data will exist somewhere without being overly redundant. ***Be sure to migrate data from physical storage every 3-5 years***

### Storage Media

Successful preservation depends in great part on storage media that are in good physical and operational condition. There are many different kinds of storage media that can be used to save and back-up digital files; each has pros and cons. They include:

### 3. Data Storage & Backup

- Desktop and Laptop Computers
- Campus servers
- Commercial Cloud Backup (e.g., Dropbox, Google Drive, Amazon, Carbonite, Spideroak). ***Caution: if Google Drive or Dropbox are set up for file-syncing, then they are not a true backup.***
- External hard drives (range from disks to USB drives)
- Optical Storage (CD, DVD)
- Paper (e.g., printouts of spreadsheets and text files)

**All storage media, whether hard drives, discs or data tapes, will wear out over time, rendering your data files inaccessible.** To ensure ongoing access to both your active data files and your data archives, it is important to continually monitor the condition of your storage media and track its age. Older storage media and media that show signs of wear should be replaced immediately. **Use the following guidelines to ensure the ongoing integrity and accessibility of your data:**

1. **Test Your Storage Media Regularly:** It is important to routinely perform test retrievals or restorations of data you are storing for extended periods on hard drives, discs or tapes. It is recommended that storage media that is used infrequently be tested at least once a year to ensure the data is accessible.
2. **Beware of Early Hardware Failures:** A certain percentage of storage media will fail early due to manufacturing defects. In particular, hard drives, thumb drives and data tapes that have electronic or moving parts can be susceptible to early failure. When putting a new drive or tape into service, it is advisable to maintain a redundant copy of your data for 30 days until the new device “settles in.”
3. **Determine the Life of Your Hard Drives:** When purchasing a new drive unit, note the Mean Time Between Failure (MTBF) of the device, which should be listed on its specifications sheet (device specifications are usually packaged with the unit, or available online). The MTBF is expressed in the number of hours on average that a device can be used before it is expected to fail. Use the MTBF to calculate how long the device can be used before it needs to be replaced, and note that date on your calendar (For example, if the MTBF of a new hard drive is 2,500 hours and you anticipate having the unit powered on for 8 hours a day during the work week, the device should last about 2 years before it needs to be replaced).
4. **Routinely Inspect and Replace Data Discs:** Contemporary CD and DVD discs are generally robust storage media that will fail more often from mishandling and improper storage than from deterioration. However lower quality discs can suffer from delamination (separation of the disc layers) or oxidation. It is advisable to inspect discs every year to detect early signs of wear. Immediately copy the data off of discs that appear to be warping or discolored. Data tapes are susceptible both to physical wear and poor environmental storage conditions. In general, it is advisable to move data stored on discs and tapes to new media every 2-5 years (specific estimates on media longevity are available on the web).
5. **Handle and Store Your Media With Care:** All storage media types are susceptible to damage from dust and dirt exposure, temperature extremes, exposure to intense light, water penetration (more so for tapes and drives than discs), and physical shock.

### 3. Data Storage & Backup

To help prolong its operational life, store your media in a dry environment with a comfortable and stable room temperature. Encapsulate all media in plastic during transportation. Provide cases or plastic sheaths for discs, and avoid handling them excessively.

6. **Multimedia (e.g., photo, video, audio) backup and archiving is a bit more challenging, in part because transcriptions and captioning can be important for interpretation, discovery, and accessibility.** Storage of images solely on local hard drives or servers is not recommended. Unaltered images should be preserved at the highest resolution possible. Store original images in separate locations to limit the chance of overwriting and losing the original image. There are a number of options for metadata for multimedia data, including [MPEG standards](#), [PBCore](#), and the US Library of Congress standards for [still images](#), [sound](#), and [moving images](#).
7. **Finally, consider the short- and long-term security of the originals:** specimens, samples, documents and data sheets, photographs, or other physical items. Options include:
  - Campus Office (long-term security/stability: poor)
  - Lab (long-term security/stability: moderate)
  - Museum/Herbarium (long-term security/stability: high)
  - Departmental Office ((long-term security/stability: poor/moderate)
  - Commercial storage facility (long-term security/stability: depends on funding source)
  - Home ((long-term security/stability: poor)
  - Library (long-term security/stability: high)
  - Dryad, Zenodo, OSF, or other digital repository with long-term archiving via CLOCKSS or similar (long-term security/stability: high)

### Backup Procedures

Backing up requires discipline and care to ensure nothing is missed and the file back-ups are secure. You can make this easier by doing the following:

1. **Plan ahead, and write down your plan.** A written data back-up and security plan includes such information as:
  - where backups are located
  - who can access backups and how they can be contacted
  - how often data should be backed up
  - what kind of backups are performed
  - who is responsible for performing the backups and their contact information. For large projects or projects with high-volume data streams include a person with secondary responsibility (the back-up back-up) in case the primary person responsible is unavailable
  - what hardware and software are used or recommended for performing backups
  - how and how often to check if backups have been performed successfully
  - the media are used to backup data
  - a list of any data that are *not* archived or backed up

### 3. Data Storage & Backup

2. **Backup your data at regular frequencies**, with backup strategies (e.g., full, incremental, differential) optimized for the data collection process and data type. For instance, you should (at least):
  - Back up when you complete your data collection activity
  - Back up after you edit / clean data
  - Streaming data (e.g., from data loggers) should be backed up at regularly scheduled points in the collection process
  - High-value data should be backed up at much higher frequencies (daily)
3. **Put procedures in place to make sure you follow your plan.**
  - set up calendar reminders
  - use checklists.
  - Ensure backup copies are identical to the originals (using checksums, file counts, etc.)
  - Be sure of one of the regular calendar items is to verify successful recovery from a backup copy.
  - ***Automate as much as possible.*** Automation simplifies frequent backups. but be careful with accidental file loss and that back-up copies are actually independent (e.g. Dropbox and a synced folder on your laptop are *not* independent copies).
4. **Ensure the Security of Restricted Data.**
  - All data collected by UF affiliates falls into one of the classifications defined in the [UF Data Policy](#). Both original and electronic archives must be secured; failure to do so can have serious consequences for both researchers and institutions.

### File formats for storage

**For long term preservation is it necessary to store data in file formats that will be readable in the future and by a variety of operating systems.** It is also important to provide descriptive information on these data file types and formats. This will facilitate data retrieval and reuse.

1. Document and store data using file-types that are open (ie., non-proprietary), uncompressed, unencrypted (if at all possible), and stable:
  - ASCII formatted files will be readable into the future. For tabular data use comma-separated values, `.csv`, for text use `.txt` (alternatively `.rtf`).
  - Images: TIFF (uncompressed)
  - Video: MPEG-4 (`.mp4`) or motion JPEG 2000 (`.jp2`)
  - AudioL Free Lossless Audio Codec (`.flac`)
  - Documentation: `.txt` (preferred), PDF/A (`.pdf`)
  - Structured, highly coded data: `.xml`
  - For geospatial (raster) data the following provide a stable format:
    - GeoTIFF/TIFF
    - ASCII Grid
    - Binary image files

### 3. Data Storage & Backup

- NetCDF
    - HDF or HDF-EOS
  - For image (Vector) data use the following file formats (these are mostly proprietary data formats; please be sure to document the Software Package, Version, Vendor, and native platform):
    - ARCVIEW software: store components of an ArcView shape file (.shp, .sbx, .sbn, .prj, and .dbf files)
    - ENVI – .evf (ENVI vector file)
    - ESRI Arc/Info export file (.e00)
  - **Note:** Certain file formats, such as shapefiles, can be made up of as many as 7 individual files. If one of those files is absent from the file assembly the shapefile data utility may be lost. Awareness of adherence to a particular file format standard can also be helpful for determining, for example, if a particular software package can read the data file. Awareness of whether that standard or format is open source or proprietary will also influence how and if the data file can be read.
2. If a particular software package required to read and work with the data file, you need to make sure you have a copy, a backup copy, *and a computer with the proper operating system to use the software.*
  3. For audio and image files, back up with the highest-quality file-type possible (i.e., the ‘lossless’ format rather than the ‘lossy’ format). For example, **jpeg** is lossy, meaning images saved with this file type will have less detail than images saved with the lossless TIFF format if at all possible back-up.

## Tools & Resources

### UF Library Guides

- UF Library: [Data File Types and Formats](#)
- UF Library: [Data Storage](#)
- UF IRB: [Data/Record Storage and Security](#)
- UF IT: [Data Classifications for Confidentiality & Security](#)

### Overview of ‘Checksums’ and Checksum Calculators

- [Fisher, Tim. What Is a Checksum?](#) (updated 15 April 2020).

### File Formats and Extensions

- UF library [recommended Data Formats for Preservation Purposes in the Florida Digital Archive](#)
- [What is XML?](#)
- [Common Windows File Extensions \(NMU IT\)](#)
- [Common File Extensions \(PC Net\)](#)

### 3. Data Storage & Backup

- [List of Open File Formats \(Wikipedia\)](#)
- [Common Windows File Extensions \(NMU IT\)](#)
- [Common File Extensions \(PC Net\)](#)
- [List of Open File Formats \(Wikipedia\)](#)
- [File formats-Best Practices \(Stanford Library\)](#)
- [Library of Congress](#)

#### UF-based Storage Options

- Overview of UF Cloud Data Services, aka [GatorCloud](#)
- [Free UF Dropbox for Education](#), UF Dropbox [FAQs](#), and information on how [your advisor](#) can submit a request for your UF Dropbox account
- [GStuite Allocation](#) for UF personnel
- [Free MS One Drive](#) (but also read why I suggest NOT using this [here](#))
- for very large data volumes or data data required extra security protection due to privacy restrictions, etc. seek the assistance of [UF Research Computing](#)

#### Storage and backup for very large volumes of data (multiple TB)

- [Cyverse](#)
- [Amazon Web Services \(AWS\) S3](#)
- [UF Research Computing](#)
- [Open Storage Network](#)
- [Argonne Leadership Computing Facility](#)
- [Globus](#)
- Uploading large files to Google drive: [tutorial blog post](#)

##### 3.1.0.1. Tools for File Syncing

- [MacDropAny \(macOS\)](#)
- [MultCloud](#)
- Dropbox (use [via UF](#) for research, teaching)
- Google Drive (use [via UF](#) for research, teaching)

#### Portable HDs and Backup Services

- **Remember:** Dropbox and Google drive are not true backups if set up as a file syncing tool.
- NY Times *Wirecutter* Reviews & Recommendations:
- [portable hard drives](#)
- [automated data backup services](#); their recommendation was [SpiderOak](#).



### 3.1.0.2. Tools for Automating Data Uploading During Data Collection:

- [IFTTT](#) and others

### Online multimedia repositories and tools

- [YouTube](#)
- [Vimeo](#)
- [Flickr](#)
- [Google Photos](#)

*These services have some advantages:*

- are often low-cost or free
- are open to all
- have functions for provide community commenting and tagging
- some provide support for explicit licenses and re-use
- provide some options for valuable metadata such as geolocation
- may allow for large-scale dissemination
- optimize usability and low barrier for participation

*However, they are commercial services, and hence have a number of potential drawbacks:*

- models for sustainability is profit-based
- may have limits on file size or resolution
- may have unclear access, backup, and reliability policies, be sure to review them carefully

There are specialized photo repositories designed with researchers in mind.

- [Tropy](#)
- [MorphBank](#)

*Some advantages of these are:*

- often open-source, free, developed and maintained by academic institutions
- domain-specific metadata fields and controlled vocabularies customized for expert users
- highly discoverable for those in the same domain
- they can provide assistance in curating metadata
- they can optimize scientific use cases such as vouchering, image analysis
- may provide APIs for sharing or re-use for other projects
- they are recognized as high-quality, scientific repositories

*But they too have some drawbacks*

- rely on research or institutional/federal funding
- may require high-quality multimedia, completeness of metadata, or restrict manipulation
- may not be open to all

### 3. Data Storage & Backup

- may have restrictions on bandwidth usage

#### A step back: what needs to be preserved?

To meet multiple goals for preservation, researchers should think broadly about the digital products that their project generates, preserve as many as possible, preserve all that they are required to preserve for as long as required or longer, and plan the appropriate preservation methods for each. Consider how long and how to preserve the following, taking into account **(1)** what would be necessary to reconstruct a complete data set if files downstream were lost **(2)**, how long each needs to be kept, and **(3)** the cost of and space required for preserving data for different lengths of time:

1. Raw data (written form, electronic form). *Raw data are almost always worth preserving.*
2. Tables, spreadsheets, or databases of raw observation records and measurements
3. Tables, spreadsheets, or databases of clean observation records and measurements. *If clean data can be easily or automatically re-created from raw data, consider not preserving. If quality control or analysis is time-consuming or expensive, then consider preserving the clean version.*
4. Intermediate products: partly summarized or coded data that are the input to the next step in an analysis
5. Documentation of protocols used to clean, summarize, or code data
6. Software or algorithms developed to prepare data (cleaning scripts) or perform analyses. *Algorithms and software source code cost very little to preserve*
7. Results of an analysis, which can themselves be starting points or ingredients in future analyses, e.g. distribution maps, population trends, mean measurements. *These may be particularly valuable for future discovery and also cost very little to preserve.*
8. Any data sets obtained from others that were used in data processing

#### Sources

1. DataONE Community Engagement & Outreach Working Group, DataONE (July 01, 2010) “Best Practice: Plan for effective multimedia management”. Accessed through the Data Management Skillbuilding Hub at <https://dataoneorg.github.io/Education/bestpractices/plan-for-effective> on Aug 31, 2020
2. DataONE Best Practices Working Group, DataONE (July 01, 2010) “Best Practice: Ensure integrity and accessibility when making backups of data”. Accessed through the Data Management Skillbuilding Hub at <https://dataoneorg.github.io/Education/bestpractices/ensure-integrity-and> on Aug 31, 2020
3. Cindy Parr, Heather Henkel, DataONE (Aug 30, 2011) “Best Practice: Create and document a data backup policy”. Accessed through the Data Management Skillbuilding Hub at <https://dataoneorg.github.io/Education/bestpractices/create-and-document> on Aug 31, 2020
4. Cindy Parr, Heather Henkel, Keven Comerford, DataONE (May 11, 2011) “Best Practice: Decide what data to preserve”. Accessed through the Data Management

### *3. Data Storage & Backup*

Skillbuilding Hub at <https://dataoneorg.github.io/Education/bestpractices/decide-what-data> on Aug 31, 2020

## 4. Data Organization in Spreadsheets

### 4.1. Things to remember when organizing Data in Spreadsheets

#### Make your data tidy

- Spreadsheets should be a rectangle, with only rows and columns.
- Each column is a different variable (a thing you are measuring, like ‘weight’ or ‘temperature’).
- One row per observation. Each cell has only one value.
- Column headers: Use short meaningful column names with no spaces or special characters. Don’t start column names with numbers. Record units in column headers.
- Don’t enter the same data on multiple spreadsheets: Use one for each category of data to avoid duplicated data and to simplify corrections (e.g., taxonomy).
- Never put multiple tables in a single spreadsheet.
- Avoid spreading data across multiple sheets
- *Collecting* data in tidy format makes it easier to *enter* data in tidy format.

#### Use consistent names, abbreviations/codes, and capitalization.

- Write dates as YYYYMMDD. Better still have separate columns for Year, Month, and Day.
- Excel is unable to parse dates from before 1899-12-31. Be careful if your data include a mix of dates before and after this date, then you’ll have mixed data types in one column. <sup>1</sup>
- Record zeros with a numeral ( 0), not a blank cells. For missing data use an appropriate null value indicator (e.g., NA).
- Don’t use formatting to convey information or to make your spreadsheet look pretty.
- *Remember that data format and excel defaults can vary by region.* For example, depending on the part of the world where a user is based, the default value for the decimal and thousands operator could be a , (comma) or a . (period); some regions use mm-dd for dates while others use dd-mm.

#### DO NOT EDIT OR CORRECT RAW DATA FILES!

- Once you are done with data entry, save your file in ‘read only’ format and make *all* corrections using scripting.

---

<sup>1</sup>The reason dates in Excel are so weird is that it is *accounting software*. It counts the days from a default of December 31, 1899, and thus stores July 2, 2014 as the serial number 41822. This is so one can easily calculate “days from a given date” for accounting purposes (like invoicing) by adding “date+XX days”.

#### 4. Data Organization in Spreadsheets

- Do *not* edit raw data after you have entered it in your spreadsheet!

### 4.2. Readings, Tools, & Resources

1. Broman, K. W., & Woo, K. H. (2018). Data organization in spreadsheets. *The American Statistician*, 72(1), 2-10. [\[read online\]](#)
2. Data Validation in Google Sheets: [blog post](#) and [video tutorial](#). A pdf version is available for download [here](#).
3. Why not bypass spreadsheets like Excel and use a csv editor like [Comma Chameleon][<https://comma-chameleon.io/>] instead? CC and other csv editors allow you to enter data in the same way - into cells, by adding and removing rows - and then export your file. But that's about it, which means you can't do many of the things (e.g., calculations, color in cells) that cause problems down the road.
4. More advanced users comfortable with R can also look into [Data Curator](#), with which you can create and edit tabular data from scratch or from a template, open Microsoft Excel and CSV files, and automatically correct common problems found in these and other file types.
5. DataONE Community Engagement & Outreach Working Group (2017) "Data Quality Control and Assurance". Accessed through the Data Management Skillbuilding Hub at [https://dataoneorg.github.io/Education/lessons/05\\_qaqc/index](https://dataoneorg.github.io/Education/lessons/05_qaqc/index) on Aug 31, 2020
6. DataONE Community Engagement & Outreach Working Group (2017) "Data Entry and Manipulation". Accessed through the Data Management Skillbuilding Hub at [https://dataoneorg.github.io/Education/lessons/04\\_entry/index](https://dataoneorg.github.io/Education/lessons/04_entry/index) on Aug 31, 2020
7. Chris Prener, Trevor Burrows (Eds.). [Data Carpentry: Data Organization in Spreadsheets for Social Scientists](#).
8. Peter R. Hoyt, Christie Bahlai, Tracy K. Teal (Eds.), Erin Alison Becker, Aleksandra Pawlik, Peter Hoyt, Francois Michonneau, Christie Bahlai, Toby Reiter, et al. (2019, July 5). [datacarpentry/spreadsheet-ecology-lesson](#): Data Carpentry: Data Organization in Spreadsheets for Ecologists, June 2019 (Version v2019.06.2). Zenodo. <http://doi.org/10.5281/zenodo.3269869>

## 5. Reproducible Data Cleanup with R

### 5.1. Set up an RStudio Project and install the relevant packages

1. File -> New Project
2. Name the Project as follows: `las_demo`
3. Create the following three folders:

```
data_raw
data_clean
code
```

You can create these in the folder using your operating systems “create folder” option *or* you can create within R studio using the `files` tab

4. Install libraries, load libraries, and run verify they work by running a few commands

### 5.2. Gather the Raw Data to be corrected

Save your original (aka ‘raw’ or ‘dirty’) data files in the `data_raw` folder.

#### ! Important

1. Never make changes to the original file of dirty/raw data. Never. **NEVER**. Always import the uncorrected original data file, make corrections using a script, and then save a new, clean file.
2. *Annotate your code with reminders for future you:* the `#` symbol (aka pound, hash tag) allows you to make comments in the script explaining what different sections or commands do. Annotate your scripts with lots of details on what different commands do, what is being done in each section, even links to websites you used to figure stuff out. Remember: ABA (*always be annotating*).

### 5.3. Take your data from messy to clean in these 6 steps:

1. Familiarize yourself with the data set
2. Check for potential errors. These can be structural errors (e.g., misaligned columns, duplicated rows/columns, missing values), data entry errors, or measurement errors. Decide how you will flag and deal with them.

## 5. Reproducible Data Cleanup with R

3. Decide how to deal with missing values
4. Identify ways to simplify data values (i.e., codes, abbreviations) and column headings
5. Write code to load the ‘raw’ data file, implement your corrections/changes, and save the ‘clean’ version of the data.

*Things to look out for:* Make a list of the what you think needs to be corrected and the steps necessary to identify and implement each correction. Some of the things to look out for include:

- Numeric values stored as character data types
- Factors stored as characters
- Duplicate rows
- Spelling mistakes
- inconsistent formatting (eg., codes, capitalizations)
- White spaces
- Missing data
- Zeros instead of null values
- Special characters (e.g. commas in numeric values instead of decimals)
- column headings with spaces between words or that start with numerals

Remember, *the characteristics of clean data set include:*

- Free of duplicate rows/values
- Error-free (correct misspellings, eliminate special characters)
- correct data type for analysis
- outliers identified and dealt in the correct way
- “tidy” data structure

💡 When working with multiple files, should you correct *before* or *after* combining?

We often need to combine multiple files with the same kind of data (i.e., surveys conducted in Year 1 and in Year 2, each of which are recorded in their own .csv file). Is it more efficient to correct each file first, then combine them, or combine first and *then* correct?

It depends and can vary from one project to another. Make an outline of the different steps and corrections to be made to each file and see if you can decide which is more efficient. Note that there might be different ways to do the same thing, this outline will help figure out which is best. For instance you could:

### ***Option 1***

1. Import table 1
2. Correct column headings in Table 1
3. Import table 2
4. Correct column headings in Table 2
5. Bind Table 1 and Table 2 Together

***but this is less efficient than...***

### ***Option 2***

1. Import table 1

2. Import table 2
3. Bind Table 1 and Table 2 Together
4. Correct the column headings in the Table

## 5.4. Tools & Resources

1. These introductions to R and R Studio were made by Professor Ethan White (UF-WEC). They are a good overview of some R basics.
  - [Intro to R and RStudio.](#)
  - [Navigate R and RStudio web page](#)
  - [Intro to R Packages](#)
  - [Expressions and Variables in R](#)
2. The Carpentries' R workshops (self-paced or taught in-person) are excellent, I use many of their materials in class:
  - [R for Social Scientists](#)
  - [Data Analysis and Visualization in R for Ecologists](#)
3. Software Carpentry lesson on [Project Management with R Studio](#)
4. Hadley Wickham wrote a book on using the tidyverse and the [online version is FREE](#). This is a phenomenal resource on using R to import, tidy, and visualize data.
5. [RStudio Cheat Sheets](#): help with commands for using the different `tidyverse` packages, RStudio shortcuts and tricks, help with R commands, and more. You definitely want the ones for Data Import, Work with Strings, Factors, Data Transformation, and Base R.
6. Where and How to ask for help
  - Hadley Wickham's advice on [how to write a good reproducible example](#) for getting help with R
  - [how to post good questions on StackOverflow](#)
  - The UF [R-users listserv](#) is *very* user friendly and a great place to post requests for help.
7. [Ten simple rules for biologists learning to program](#)
8. Lot's more on the course's [‘Resources’ page](#)



## 6. Additional (interesting) Reading

1. Lewis, Keith P., Eric Vander Wal, and David A. Fifield. 2018. Wildlife biology, big data, and reproducible research. *Wildlife Society Bulletin* 42(1): 172-179.
2. White EP, Baldrige E, Brym ZT, Locey KJ, McGlinn DJ, Supp SR. 2013. Nine simple ways to make it easier to (re)use your data. *Ideas in Ecology and Evolution*. 6(2):1-10.
3. [The humanities have a ‘reproducibility’ problem](#)
4. [The humanities do not need a replication drive](#)
5. [Reproducible Research: A primer for the social sciences](#)
6. [Replicability and replication in the humanities](#)
7. [Towards reproducible science in the digital humanities](#)
8. [The possibility and desirability of replication in the humanities](#)
9. [Reproducible Research: A Retrospective](#)

### *For when you feel more comfortable with R and programming*

1. Bryan, J. (2018). Excuse me, do you have a moment to talk about version control? *The American Statistician*, 72(1), 20-27.
2. Wilson, Greg, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. Good enough practices in scientific computing. *PLoS Computational Biology* 13, no. 6 (2017): e1005510.

## 7. QA/QC: Reducing Error Entry/Finding and Correcting Errors

### 7.1. 'Data Validation' Rules

1. Data Validation in Microsoft's Excel [\[link\]](#)

### 7.2. Data entry with 'Text-to-Speech'

#### MS Excel (& other MS programs)

1. Converting text to speech in Excel for Microsoft 365, Excel 2010-2021 [\[link\]](#).
2. Microsoft for Mac: Hear selected text read aloud from Excel, Word, Power Point, and Outlook [\[link\]](#). See also '*System Preferences->Accessibility*'
3. Video Tutorials
  - [Tutorial Video 1](#). This will allow you to select rows or columns of Excel and have them read back to you.
  - [Tutorial Video 2](#): it's not as thorough, but it is a bit easier to see the menu
  - [Tutorial Video 3](#). Because why not a third one?

#### Text-to-Speech in Google Docs

1. If you prefer working in Google Docs you can do the same thing. [This article](#) will show you how. You can also watch [this tutorial video](#).

#### Better still: 'speak-on-enter' to confirm data values as entered

1. Speak-on-enter [\[tutorial:\]](#)

#### Speech-to-Text for keyboardless data entry

1. Google Sheets: [Typing with your voice](#)
2. MS Word: [How to dictate documents](#)

### 7.3. Overview of MS Accessibility Tools

1. ‘Narrator’ function [link](#)

### 7.4. R Packages for Data QA/QC

1. [janitor](#): simple functions for examining and cleaning dirty data. It was built with beginning and intermediate R users in mind and is optimized for user-friendliness. Advanced R users can already do everything covered here, but with janitor they can do it faster. (see this [mini-tutorial](#))
  - clean column names
  - remove empty rows and columns
  - remove duplicated rows
2. [cleanr](#). small R package for cleaning and checking data columns in a fast and easy way.
3. [unheadr](#): used to wrangle spreadsheets with embedded subheaders or values wrapped accross several rows (highlighting, merged cells, etc).

#### 7.4.1. For more advanced R users:

1. Richard’s Iannone [pointblank](#) R package for Data Validation and Organization of Metadata [link](#)
2. [validate](#): designed to test data against a reusable set of data validation rules, investigate, summarize, and visualize data validation results, among other things
3. [Data Curator](#) is a simple desktop data editor to help describe, validate and share usable open data.
4. Kim, A. Y., Herrmann, V., Barreto, R., Calkins, B., Gonzalez-Akre, E., Johnson, D. J., Jordan, J. A., Magee, L., McGregor, I. R., Montero, N., Novak, K., Rogers, T., Shue, J., & Anderson-Teixeira, K. J. (2022). Implementing GitHub Actions continuous integration to reduce error rates in ecological data collection. *Methods in Ecology and Evolution*, 13, 2572–2585. <https://doi.org/10.1111/2041-210X.13982>

### 7.5. Readings

1. Campbell, J. L. *et al.* 2013. Quantity is nothing without quality: automated QA/QC for streaming environmental sensor data. *BioScience*, 63(7): 574-585. [link](#)

For more advanced users of R/Github

1. Kim, A. Y. *et al.* 2022. Implementing GitHub Actions continuous integration to reduce error rates in ecological data collection. *Methods in Ecology and Evolution*, 13, 2572–2585. <https://doi.org/10.1111/2041-210X.13982>

## 8. Sources

DataONE Education Module: Data Quality Control and Assurance. DataONE. Retrieved Nov12, 2012. From [http://www.dataone.org/sites/all/documents/L05\\_DataQualityControlAssurance.pptx](http://www.dataone.org/sites/all/documents/L05_DataQualityControlAssurance.pptx)

Barchard, K. A., & Pace, L. A. (2011). Preventing human error: The impact of data entry methods on data accuracy and statistical results. *Computers in Human Behavior*, 27(5), 1834-1839. [link](#)

Atkinson, I. (2012). Accuracy of data transfer: double data entry and estimating levels of error. *Journal of Clinical Nursing*, 21(19pt20), 2730-2735.[link](#)

Goldberg, S. I., Niemierko, A., & Turchin, A. (2008). Analysis of data errors in clinical research databases. *AMIA Annual Symposium proceedings. AMIA Symposium*, 2008, 242-246. [link](#)

**Part II.**

**Assignments**

## 9. Assignments Overview

Grades in the course will be based on the following assignments:

### 9.1. Weekly In-class activities

- **Overview:** Most of the in-class assignments involve hands-on practice with data collection or manipulation. In some weeks, however, assignment will be the submission of questions for group discussion or brief reflection on the issues from the readings. Most in-class assignments are designed to be completed during the class session, but to ensure students master the concepts rather than rush through them *they can be submitted anytime until 9 am the following Friday.*

### 9.2. Reproducible Data Organization Project

- **Overview:** This project is an opportunity to put some of the lessons learned into practice with a data set of your own. Your assignment is to **(1)** clean and organize a ‘messy’ data set and prepare metadata describing the resulting ‘clean’ data. The complete project requires the submission of these three items via the course Canvas website:
  - (1) R code that imports, cleans, and organizes, and saves ‘messy’ data
  - (2) The resulting corrected and organized data in an appropriate format
  - (3) Metadata describing the corrected data set

### 9.3. Thesis Research DMP

- **Overview:** The Data Management Plan (DMP) is a critical document describing the data to be collected for a research project, how it will be stored and managed, and the investigator with primary responsibility for its management. Many funding agencies, including NSF and NIH, now require a DMP with all grant applications. Each student will prepare a Data Management Plan (DMP) for their thesis research.

**Part III.**

**Useful Resources**

# 10. Useful resources for Data Collection & Managment

## 10.1. R Programming

### Essential Resources

1. ***R for Data Science* book:** Hadley Wickham wrote a book on using the tidyverse and *the online version is FREE*. This is a phenomenal resource on using R to import, tidy, and visualize data.
2. **Posit Cheat Sheets:** help with commands for using the different tidyverse packages, RStudio shortcuts and tricks, help with R commands, and more. You definitely want the ones for Data Import, Work with Strings, Factors, Data Transformation, and Base R.
3. **RStudio Keyboard Shortcuts:** A list of the keyboard shortcuts for Mac, Windows, and Linux can be found [here](#).
4. **Where and How to ask for help:**
  - Hadley Wickham's advice on [how to write a good reproducible example](#) for getting help with R
  - [how to post good questions on StackOverflow](#)
  - The UF [R-users listserv](#) is *very* user friendly and a great place to post requests for help.

### Tutorials

1. [Paul van der Laken's List](#) of books, tutorials, and other resources on topics ranging from data manipulation to data validation to data visualization.
2. [R Essential Training: Wrangling and Visualizing Data](#). (requires a UF email address to access LinkedIn Learning).
3. Software Carpentry: [Using RStudio for Project Organization & Management](#)
4. [Swirl](#): learn R programming interactively, at your own pace, and in the R console.
5. [R Bootcamp](#) by Ted Laderas and Jessica Minnier. Learn R in your browser.
6. [How to clean messy data in R](#)



7. [The Ultimate Guide to Data Cleaning](#) is written for Python users but the principles apply regardless of language.

## 10.2. Specific Data Cleaning and Management Problems

### Dates & Times

1. [Handling dates and times in R](#)

### Text & Text Mining

2. Text Mining: [tidytext](#) package

### Qualtrics

3. Working with Qualtrics survey data: [qualtRics](#) package

### Text Extraction

4. **Optical Character Recognition (OCR):** extract text from images: [tesseract](#) package
5. Extract text & metadata from pdf files: [pdftools](#) package

### Images & Image Processing

6. Image processing: the [magick](#) package

## 10.3. Advanced R Packages for Data Management

1. [DataCurator](#) package: ‘a simple desktop data editor to help describe, validate and share usable open data’.
2. [RegExr](#): online tool to learn, build, & test Regular Expressions (RegEx / RegExp)
3. [janitor](#) (cleanup of file names, etc.)
4. [ROpenSci](#): tools for accessing, manipulating, and visualizing open data

## 10.4. Data Visualization

1. [Data Visualization: a practical introduction](#) by Kieran Healy is my favorite introductory (yet super-comprehensive) book on data visualization with R. If you scroll down to the bottom of the page you can download the datasets and code used to make the figures in the book, which makes life much easier.

## 10.5. Slide & Presentations

1. Make [slide presentations with R](#)

## 10.6. Documents & Reports

1. [knitr](#) overview: reproducible documents with R

## 10.7. Discipline-specific R Resources

### History

1. [historydata](#) package: Sample data sets for historians learning R. They include population, institutional, religious, military, and prosopographical data suitable for mapping, quantitative analysis, and network analysis.
2. [The Programming Historian](#) Website: wide range of topics, from text analysis to OpenRefine

### Psychology

1. [‘Programming for Psychologists: Data Creation and Analysis’](#) by Matthew J. C. Crump

## 10.8. Data Archives

1. [Qualitative Data Repository](#): dedicated archive for storing and sharing digital data (and accompanying documentation) generated or collected through qualitative and multi-method research in the social sciences and related disciplines.
2. [Data Dryad](#): open data publishing platform and a community committed to the open availability and routine re-use of all research data.
3. [ICPSR](#): data access, curation, and analytical methods for social science.

## **11. Slide decks**

## 12. Slides for different course topics

- [Introduction](#)
- [File Names](#)
- [‘Reproducible’ Data Management](#)
- Slides - Introduction
- Slides - File Names
- Slides - ‘Reproducible’ Data Management